Parameters

$g = 9.81 m/s^2$

$L = 0.15 m$

$m = 0.1 kg$

$\omega_1 = 0.1 m$

Initial Angle $= 30°$

Pendulum is released from rest

Simulation runs for 11 seconds

Make 2 1st-Order ODEs to solve this

$\dot{\theta} = \omega$

$\dot{\omega} = f(\theta, \omega, \Omega)$

$\dot{\theta} = \omega$

$\dot{\omega} = -g/L \sin\theta + \Omega^2/L(\omega_1 + L\sin\theta)\cos\theta$

The natural frequency for a pendulum $= -\sqrt{g/L} = 8.1$ rad/s, so our slow speed should be below 8.1 rad/s, and our fast speed should be above 8.1 rad/s

This is why we decided to go with:

0.3 rad/s = slow speed

14.0 rad/s = fast speed

Scanned with CamScanner